An elementary VSL lane-drop task: model and sensitivity

Revisions responding to feedback 11 are shown in red.

1. Research question

This elementary variable-speed-limit (VSL) task separates three questions:

- **Information:** Which omitted lane-loading information changes merge service when ending-lane traffic and total continuing traffic agree?
- **Decision:** Can retaining that information change the useful common-speed advice after both travel and all waiting time are counted?
- **Gradient:** Does the complete travel-time sensitivity predict beneficial speed changes, checked against actual objective changes and equal-information optimizer references?

2. Task and global assumptions

Concrete task. Control one permanent three-to-two lane drop for one specified finite cohort, using one constant common speed advice. The road has ending lane E, adjacent continuing lane A and far continuing lane B. Preparation ends at L_p; a terminal of length b ends at the choke $L = L_p + b$. At time zero, independent lane densities $(e_0, a_0, b_0) \geq 0$ are uniform on the common footprint $[0, \ell]$, $0 < \ell \leq L_p$. Set $r = e_0 + a_0$ and total mass $M = \ell(e_0 + a_0 + b_0) > 0$. The terminal and recovery history start empty; there are no later external arrivals. Choose one constant common advice $v \in [v_-, v_+]$, $0 < v_- < v_+$, with fixed initial state, geometry and physical parameters. The objective counts every initial unit's time until it crosses the choke.

Global modeling assumptions. These specify the traffic represented by this task, separately from the sufficient operating bounds below.

- **MA1: continuous, homogeneous traffic.** Each lane carries continuous vehicle-equivalent mass. The remaining preparation body retains lane densities without internal longitudinal variation or individual maneuvers.
- **MA2: advice acts through traffic state.** Unblocked traffic follows the common advice. Backward-wave speed w, jam density k_j, geometry and physical interaction/recovery coefficients stay fixed as advice varies; merge service responds to the resulting local traffic state.
- **MA3: two-stage lane coupling.** E can transfer into A before the terminal; successful early transfer becomes ordinary A. Residual E traverses the terminal without ordinary lane changes and merges at the choke. B has separate service; no other lateral exchange is represented.
- **MA4: point-service closure.** Current A/E conditions determine priority/cooperation, maneuver work and representative boundary speeds. Those speeds describe free or queued service, rather than individual trajectories.
- **MA5: recovery memory.** Only forced E discharge generates recovery loss. One shared A/E budget depends on the retained ages and insertion speeds of the most recent fixed amount of E mass.

Triangular-diagram and cumulative-boundary ingredients follow the link transmission model (LTM) [1]; the advice-response family is selected here, with algebraic capacity/inverse maps:

$$K = wk_j, \quad k_c(v) = \frac{K}{w+v}, \quad C(v) = vk_c(v), \qquad c(u) = \frac{Ku}{w+u}, \quad U(g) = \frac{wg}{K-g}. \tag{1}$$

Here k_c is critical density, C capacity, $c(u)$ the congested-branch flow associated with speed u, and U its inverse. R and R_B are undisturbed A/E and B service ceilings, respectively. Use positive physical scales and the sufficient operating domain

$$e_0, a_0, b_0 \leq k_c(v_+), \qquad v_+ r < R \leq C(v_-), \qquad v_+ b_0 \leq R_B \leq C(v_-), \qquad M < k_j b. \tag{2}$$

It keeps preparation free, terminal receiving available and B unqueued. Work and recovery choices are specified below; $[x]_+ = \max(x, 0)$.

Why this test. Lane-based conservation and finite receiving constraints [2], [3], together with insertion-speed effects on discharge [4], [5], motivate conserved, feasible traffic and state-sensitive merge service. Strong aggregate VSL models already retain traffic history and capacity drop [6], but do not explicitly resolve the A/B split. Lane-resolved models [2], [3] supply that state; their traffic predictions alone do not answer this task's matched-information control/gradient questions. We therefore require complete travel-time accounting through clearance, lawful history inference, and separate checks of information value and equal-information gradient usefulness, including a sufficient compact peer and scalar search.

3. Traffic model

Picture E traffic approaching the lane end. Some finds room in A early and continues as ordinary A traffic. What remains in E crosses a short terminal and must merge at the choke, competing with A for passage. A slow forced entry can keep restricting following traffic even after it has crossed the completion line; the restriction fades as downstream traffic accelerates. The four components below represent these early opportunities, travel to the merging point, competition for passage, and the lingering recovery effect.

3.1. Pre-choke merging. Let $e(t), a(t)$ be the remaining preparation body's E/A densities. Own A flow uses receiving first. E transfer is limited by its donor offer and by remaining adjacent service. Paired conservative sources follow [2]; receiving work and own-lane priority draw on [3]. The homogeneous minimum and interaction length below are defined here:

$$S(v,a) = \min\{C(v), w(k_j - a)\}, \qquad f = \min\left\{ve, \frac{S(v,a) - va}{\gamma}\right\}, \qquad \dot{e} = -f/h, \quad \dot{a} = f/h. \tag{3}$$

Here $h > 0$ is a physical interaction length, $\gamma \geq 1$ is a receiving-work factor, and f/h is the lateral density source. Thus $e + a = r$ and b_0 stays constant. Donor-limited decay is exponential per traveled distance; advice affects preparation through normalized receiving headroom $k_c(v)$.

3.2. Finite terminal and the merging point. The terminal separates reaching the end of preparation from being ready to pass the choke: traffic first travels distance b, then waits only if point service cannot accept it. For $j \in \{E, A, B\}$, let F_j, P_j, Y_j be cumulative terminal entry, service-eligible mass and departure. The initially empty count histories represent only this cohort. Boundary travel follows LTM [1]; the preparation handoff is defined here. The cohort enters during $(L_p - \ell)/v < t < L_p/v$:

$$\dot{F}_E = ve(t), \quad \dot{F}_A = v[r - e(t)], \quad \dot{F}_B = vb_0; \qquad P_j(t) = F_j(t - b/v). \tag{4}$$

Define physical terminal stock $N_j = F_j - Y_j$, eligible waiting mass $q_j = P_j - Y_j$, offer $\lambda_j = \dot{P}_j$ and discharge $g_j = \dot{Y}_j$. Departures satisfy $0 \leq Y_j \leq P_j \leq F_j$, with $\dot{q}_j = \lambda_j - g_j$. Domain (2) protects finite storage and gives $Y_B = P_B$. Early-merged E becomes ordinary A; residual E travels through the terminal and merges at the choke.

3.3. Forced merging and service sharing. The point has a shared work budget $\mathcal{B}$, measured as A-equivalent service per unit time. A costs one work unit per mass; E's maneuver costs more when A is faster. For lane $j = A, E$, u_j is the model's representative speed at the merging boundary: $u_j = v$ for unqueued traffic and $u_j = U(g_j)$ for queued traffic. The queued relation is an equilibrium closure, not an individual speed trajectory. Receiving-work accounting is adapted from [3]; the specific speed-mismatch work and allocation below are defined here:

$$\begin{aligned} c_E &= c_0 + \chi g_A [u_A - u_E]_+, \qquad g_A + c_E g_E \le \mathcal{B}, \\ H(g; g_A, u_A) &= g \cdot \big(c_0 + \chi g_A [u_A - U(g)]_+\big) = x. \end{aligned} \tag{5}$$

$c_0 \ge 1$ is baseline E work and $\beta \ge 0$ sets mismatch strength. Set $\chi = \beta/[C(v_+)v_+]$ once and hold it fixed. Require $c_0 > \beta(1 + v_+/w)$ so each feasible root $H(g) = x$ is unique. When E is queued, **solve for** $g = g_E$: A's flow g_A, speed u_A and allocated E work x come from the sharing table. H is the fully specified work consumed by a candidate E flow g; the dot denotes ordinary multiplication of that flow by its work per unit mass. A fitting free A offer has priority; queued A cooperates through equal work and unused-work redistribution:

A, E status	Selected service and free-admission condition
Free, free	$g_A = \lambda_A,\ g_E = \lambda_E$ if $\lambda_A + c_0\lambda_E \le \mathcal{B}$.
Free, queued	$g_A = \lambda_A \le \mathcal{B}, \quad H(g_E; \lambda_A, v) = \mathcal{B} - \lambda_A$.
Queued, queued	$g_A = \mathcal{B}/2, \quad H(g_E; \mathcal{B}/2, U(\mathcal{B}/2)) = \mathcal{B}/2$.
Queued, free	$g_E = \lambda_E,\ g_A = \mathcal{B} - c_0\lambda_E$ if $c_0\lambda_E \le \mathcal{B}/2$.

A queue starts only when arriving traffic strictly exceeds feasible service. When all waiting traffic has passed, the lane becomes unqueued if its current arrivals can pass immediately; equality permits that continuation. After an arrival-rate change, service follows the new arrivals immediately. Recovery is a separate physical aftereffect: clearing a queue does not erase the influence of traffic already discharged. These transition rules and the sharing table are defined here.

3.4. Recovery and the shared budget. Crossing the choke completes an E entry's own trip, but a slow insertion can still hold back the A/E traffic waiting behind it. Downstream acceleration gradually releases that restriction. Point-merge studies supply this mechanism [4], [5]. For one entry with insertion speed u, let $\theta \ge 0$ be time since crossing and $a_r > 0$ the recovery acceleration. Differentiating the isolated cumulative passing expression in [5] gives $c(V)$; capping against undisturbed service R defines the loss below:

$$V(\theta; u) = \sqrt{(w+u)^2 + 2wa_r\theta} - w, \qquad D(\theta; u) = [R - c(V(\theta; u))]_+. \tag{6}$$

V summarizes how the downstream acceleration and backward-wave propagation restore service at the choke. It is an equivalent speed in the boundary-flow map c, not the discharged vehicle's actual speed. D is the amount of the undisturbed A/E service R still unavailable because of this insertion; it vanishes once $c(V)$ reaches R. To combine entries, ξ labels cumulative discharged E mass, $T(\xi)$ is that mass's choke-crossing time, and $u_E(T(\xi))$ its insertion speed. Set $m = 1$ vehicle-equivalent. Our recent-mass rule averages the remaining losses over at most the latest m E mass and subtracts the reading p_{rec} from R to obtain the shared budget:

$$p_{\text{rec}}(t) = \frac{1}{m}\int_{[Y_E(t)-m]_+}^{Y_E(t)} D\big(t - T(\xi); u_E(T(\xi))\big)\,\mathrm{d}\xi, \qquad \mathcal{B}(t) = [R - p_{\text{rec}}(t)]_+. \tag{7}$$

All retained profiles recover with age. Once m has passed, newer E replaces the oldest retained mass; the denominator remains m even before fill. A consumes the shared budget but adds no separate loss profile. The capped single-entry profile is derived/adapted from [5]; the E-only source and recent-mass renewal are defined here.

4. How the task tests information

Compare the same initial E mass and continuing total with different A/B splits. A busier A can restrict preparation, leaving more E for forced work and recovery. Advice trades these effects against ordinary travel time. With a fixed horizon H_* beyond cohort clearance, minimize complete total system time (TST):

$$J(v) = \int_0^{H_*} \left[M - \sum_j Y_j(t)\right] \mathrm{d}t = \frac{M(L - \ell/2)}{v} + \int_0^{H_*} (q_A + q_E)\,\mathrm{d}t. \tag{8}$$

The predictors share geometry and physical parameters and may use all inference from admitted predecision histories: aggregate density, flow, speed and cumulative counts, including any permitted E-resolved history. One knows initial E and combined A+B mass; the other knows each lane. An equal-information compact peer adds just initial A mass to the former, reconstructing all three initial densities and the same dynamics. A separate optimizer comparison uses the same information and model for gradient-based advice and bounded scalar search.

5. Complete sensitivity

The model-specific sensitivities are derived here at fixed initial traffic and physical parameters. Advice changes preparation and arrival timing; these change passage, insertion speeds, recovery and waiting. Differentiate that whole path, including when queues form/clear and old recovery entries leave the account. Here H is the scalar service-work function defined in Section 3.3, Eq. (5). The terms H_g, H_{g_A} and H_{u_A} are its first partial derivatives with respect to E discharge, A discharge and A's representative boundary speed, respectively, evaluated at the current service solution. Each measures how required work changes with one argument while the others are held fixed. A subscript v denotes advice sensitivity. On smooth branches,

$$g_{E,v} = \frac{x_v - H_{g_A} g_{A,v} - H_{u_A} u_{A,v}}{H_g}, \qquad H_g > 0. \tag{9}$$

Let s_m be the crossing time of the oldest retained E mass. For a filled window with $g_E(s_m) > 0$, $Y_E(s_m) = Y_E(t) - m$; write $Z_E = Y_{E,v}$. The cutoff relation below is derived here. Differentiate (7) including insertion speeds, flows, retirement and source events. A branch switch at $t = \tau$ satisfies a condition $G(t, v) = 0$ along the predicted traffic/history. For an isolated crossing with $G_t^- \neq 0$, the remaining event corrections follow [7]. Here X collects continuous traffic coordinates and $-/+$ mean just before/after the switch:

$$\begin{gathered} g_E(s_m) s_{m,v} = Z_E(t) - Z_E(s_m), \qquad \tau_v = -G_v^- / G_t^-, \\ X_v^+ = X_v^- + (\dot{X}^- - \dot{X}^+)\tau_v. \end{gathered} \tag{10}$$

Thus the complete derivative, on a fixed regular event/history sequence, is

$$J'(v) = -\int_0^{H_*} \sum_j Y_{j,v}(t)\,\mathrm{d}t = -\frac{M(L - \ell/2)}{v^2} + \int_0^{H_*} (q_{A,v} + q_{E,v})\,\mathrm{d}t. \tag{11}$$

References

[1] W.-L. Jin, "Continuous formulations and analytical properties of the link transmission model," *Transp. Res. B, Methodol.*, vol. 74, pp. 88–103, 2015, doi: 10.1016/j.trb.2014.12.006.

[2] H. H. S. Nagalur Subraveti, V. L. Knoop, and B. van Arem, "First order multi-lane traffic flow model—an incentive based macroscopic model to represent lane change dynamics," *Transportmetrica B, Transp. Dyn.*, vol. 7, no. 1, pp. 1758–1779, 2019, doi: 10.1080/21680566.2019.1700846.

[3] M. Carey, C. Balijepalli, and D. Watling, "Extending the cell transmission model to multiple lanes and lane-changing," *Netw. Spat. Econ.*, vol. 15, pp. 507–535, 2015, doi: 10.1007/s11067-013-9193-7.

[4] L. Leclercq, J. A. Laval, and N. Chiabaut, "Capacity drops at merges: An endogenous model," *Procedia Soc. Behav. Sci.*, vol. 17, pp. 12–26, 2011, doi: 10.1016/j.sbspro.2011.04.505.

[5] L. Leclercq, V. L. Knoop, F. Marczak, and S. P. Hoogendoorn, "Capacity drops at merges: New analytical investigations," *Transp. Res. C, Emerg. Technol.*, vol. 62, pp. 171–181, 2016, doi: 10.1016/j.trc.2015.06.025.

[6] L. Wei, Y. Han, and M. Wang, "A first-order link-based flow model with variable speed limits and capacity drops for freeway networks," *Comput.-Aided Civ. Infrastruct. Eng.*, vol. 40, no. 21, pp. 3200–3217, 2025, doi: 10.1111/mice.13492.

[7] N. J. Kong, J. J. Payne, J. Zhu, and A. M. Johnson, "Saltation matrices: The essential tool for linearizing hybrid dynamical systems," *Proc. IEEE*, vol. 112, no. 6, pp. 585–608, Jun. 2024, doi: 10.1109/JPROC.2024.3440211.